

SOLIDWORKS API (.NET VBA C#) — Reference

(156/337): IMaterialVisualPropertiesData- IMathTransform

SOLIDWORKS API Documentation (converted from CHM)

IMaterialVisualPropertiesData (Type)

IMaterialVisualPropertiesData Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : IMaterialVisualPropertiesData Interface

Allows access to a material on a part.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface IMaterialVisualPropertiesData
```

Visual Basic (Usage)



```
Dim instance As IMaterialVisualPropertiesData
```

C#

```
public interface IMaterialVisualPropertiesData
```

C++/CLI

```
public interface class IMaterialVisualPropertiesData
```

Visual Basic for Applications (VBA) Syntax

See [MaterialVisualPropertiesData](#).

Example

[Get and Set Material Visual Properties \(VBA\)](#)
[Get and Set Material Visual Properties \(VB.NET\)](#)
[Get and Set Material Visual Properties \(C#\)](#)

Accessors

[IPartDoc::GetMaterialVisualProperties](#)

Access Diagram

[MaterialVisualPropertiesData](#)

See Also

[IMaterialVisualPropertiesData Members](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IPartDoc Interface](#)

IMaterialVisualPropertiesData Interface Members

[See Also](#) [Properties](#)



[SolidWorks.Interop.sldworks Namespace](#) : IMaterialVisualPropertiesData Interface

The following tables list the members exposed by [IMaterialVisualPropertiesData](#).

Public Properties

Name	Description
 AdvancedGraphics	Obsolete. Not superseded.
 Angle	Gets or sets the angle of the texture.

 ApplyAppearance	Gets or sets whether to apply the appearance of material.
 ApplyMaterialColorToPart	Gets or sets whether the part color matches the material color.
 ApplyMaterialHatchToSection	Gets or sets whether the drawing section views use the material hatch.
 BlendColor	Gets or sets whether to blend the part color with the texture.
 RealView	Gets or sets whether textures are displayed in RealView or Standard graphics.
 Scale2	Gets or sets the scale factor for the standard texture.

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See Also

[IMaterialVisualPropertiesData Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IPartDoc Interface](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IMaterialVisualPropertiesData Interface Properties

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : [IMaterialVisualPropertiesData Interface](#)

For a list of all members of this type, see [IMaterialVisualPropertiesData members](#).

Public Properties

Name	Description
 AdvancedGraphics	Obsolete. Not superseded.
 Angle	Gets or sets the angle of the texture.
 ApplyAppearance	Gets or sets whether to apply the appearance of material.
 ApplyMaterialColorToPart	Gets or sets whether the part color matches the material color.
 ApplyMaterialHatchToSection	Gets or sets whether the drawing section views use the material hatch.
 BlendColor	Gets or sets whether to blend the part color with the texture.
 RealView	Gets or sets whether textures are displayed in RealView or Standard graphics.
 Scale2	Gets or sets the scale factor for the standard texture.

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See Also

[IMaterialVisualPropertiesData Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IPartDoc Interface](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

AdvancedGraphics Property (IMaterialVisualPropertiesData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMaterialVisualPropertiesData Interface](#) : [AdvancedGraphics Property \(IMaterialVisualPropertiesData\)](#)

Obsolete. Not superseded.

.NET Syntax

Visual Basic (Declaration)

Property AdvancedGraphics As System.Boolean

Visual Basic (Usage)



```
Dim instance As IMaterialVisualPropertiesData
Dim value As System.Boolean

instance.AdvancedGraphics = value
```

```
value = instance.AdvancedGraphics
```

C#

```
System.bool AdvancedGraphics {get; set;}
```

C++/CLI

```
property System.bool AdvancedGraphics {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True if the part is displayed with shaded textures, false if not

Visual Basic for Applications (VBA) Syntax

See [MaterialVisualPropertiesData::AdvancedGraphics](#).

See Also

[IMaterialVisualPropertiesData Interface](#)
[IMaterialVisualPropertiesData Members](#)
[IMaterialVisualPropertiesData::RealView Property](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

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Angle Property (IMaterialVisualPropertiesData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMaterialVisualPropertiesData Interface](#) : Angle Property (IMaterialVisualPropertiesData)

Gets or sets the angle of the texture.

.NET Syntax

Visual Basic (Declaration)

```
Property Angle As System.Double
```

Visual Basic (Usage)



```
Dim instance As IMaterialVisualPropertiesData  
Dim value As System.Double  
  
instance.Angle = value  
value = instance.Angle
```

C#

```
System.double Angle {get; set;}
```

C++/CLI

```
property System.double Angle {  
    System.double get();  
    void set ( & System.double value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Angle in radians

Visual Basic for Applications (VBA) Syntax

See [MaterialVisualPropertiesData::Angle](#).

Example

See [IMaterialVisualPropertiesData](#) examples.

Remarks

This property only applies to SOLIDWORKS standard textures.

See Also

[IMaterialVisualPropertiesData Interface](#)
[IMaterialVisualPropertiesData Members](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

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ApplyAppearance Property (IMaterialVisualPropertiesData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMaterialVisualPropertiesData Interface](#) : ApplyAppearance Property (IMaterialVisualPropertiesData)

Gets or sets whether to apply the appearance of material.

.NET Syntax

Visual Basic (Declaration)

Property ApplyAppearance As System.Boolean

Visual Basic (Usage)



```
Dim instance As IMaterialVisualPropertiesData  
Dim value As System.Boolean
```

```
instance.ApplyAppearance = value
```

```
value = instance.ApplyAppearance
```

C#

```
System.bool ApplyAppearance {get; set;}
```

C++/CLI

```
property System.bool ApplyAppearance {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to apply the appearance of material, false to not (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [MaterialVisualPropertiesData::ApplyAppearance](#).

Example

See [IMaterialVisualPropertiesData](#) examples.

Remarks

If this property is...	Then...
False	There is no change in the appearance of the body when material is changed.
True	The appearance of the body changes when material is changed.

See Also

[IMaterialVisualPropertiesData Interface](#)
[IMaterialVisualPropertiesData Members](#)

Availability

SOLIDWORKS 2013 FCS, Revision Number 21.0

SOLIDWORKS API Help

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ApplyMaterialColorToPart Property (IMaterialVisualPropertiesData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMaterialVisualPropertiesData Interface](#) : ApplyMaterialColorToPart Property (IMaterialVisualPropertiesData)

Gets or sets whether the part color matches the material color.

.NET Syntax

Visual Basic (Declaration)

```
Property ApplyMaterialColorToPart As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IMaterialVisualPropertiesData  
Dim value As System.Boolean
```

```
instance.ApplyMaterialColorToPart = value
```

```
value = instance.ApplyMaterialColorToPart
```

C#

```
System.bool ApplyMaterialColorToPart {get; set;}
```

C++/CLI

```
property System.bool ApplyMaterialColorToPart {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True if the part color matches the material color, false if not

Visual Basic for Applications (VBA) Syntax

See [MaterialVisualPropertiesData::ApplyMaterialColorToPart](#).

Example

See [IMaterialVisualPropertiesData](#) examples.

See Also

[IMaterialVisualPropertiesData Interface](#)
[IMaterialVisualPropertiesData Members](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

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ApplyMaterialHatchToSection Property (IMaterialVisualPropertiesData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMaterialVisualPropertiesData Interface](#) : ApplyMaterialHatchToSection Property (IMaterialVisualPropertiesData)

Gets or sets whether the drawing section views use the material hatch.

.NET Syntax

Visual Basic (Declaration)

Property ApplyMaterialHatchToSection As System.Boolean

Visual Basic (Usage)



```
Dim instance As IMaterialVisualPropertiesData
Dim value As System.Boolean
```

```
instance.ApplyMaterialHatchToSection = value
```

```
value = instance.ApplyMaterialHatchToSection
```

C#

```
System.bool ApplyMaterialHatchToSection {get; set;}
```

C++/CLI

```
property System.bool ApplyMaterialHatchToSection {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True if the [drawing section views](#) use the material hatch, false if not

Visual Basic for Applications (VBA) Syntax

See [MaterialVisualPropertiesData::ApplyMaterialHatchToSection](#).

Example

See [IMaterialVisualPropertiesData](#) examples.

See Also

[IMaterialVisualPropertiesData Interface](#)
[IMaterialVisualPropertiesData Members](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

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BlendColor Property (IMaterialVisualPropertiesData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMaterialVisualPropertiesData Interface](#) : BlendColor Property (IMaterialVisualPropertiesData)

Gets or sets whether to blend the part color with the texture.

Visual Basic Syntax

Visual Basic (Declaration)

Property BlendColor As System.Boolean

Visual Basic (Usage)



```
Dim instance As IMaterialVisualPropertiesData  
Dim value As System.Boolean  
  
instance.BlendColor = value  
value = instance.BlendColor
```

C#

```
System.bool BlendColor {get; set;}
```

C++/CLI

```
property System.bool BlendColor {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to blend the color with the texture, false to not

Visual Basic for Applications (VBA) Syntax

See [MaterialVisualPropertiesData::BlendColor](#).

Example

See [IMaterialVisualPropertiesData](#) examples.

Remarks

This property on applies to SOLIDWORKS standard textures.

See Also

[IMaterialVisualPropertiesData Interface](#)
[IMaterialVisualPropertiesData Members](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

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RealView Property (IMaterialVisualPropertiesData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMaterialVisualPropertiesData Interface](#) : RealView Property (IMaterialVisualPropertiesData)

Gets or sets whether textures are displayed in RealView or Standard graphics.

Visual Basic Syntax

Visual Basic (Declaration)

Property RealView As System.Boolean

Visual Basic (Usage)




```
Dim instance As IMaterialVisualPropertiesData
Dim value As System.Boolean

instance.RealView = value

value = instance.RealView
```

C#

```
System.bool RealView {get; set;}
```

C++/CLI

```
property System.bool RealView {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True if textures are RealView graphics, false if Standard graphics

Visual Basic for Applications (VBA) Syntax

See [MaterialVisualPropertiesData.RealView](#).

Example

See [IMaterialVisualPropertiesData](#) examples.

Remarks

[IMaterialVisualPropertiesData.AdvancedGraphics](#) must be True to set this property.

See Also

[IMaterialVisualPropertiesData Interface](#)
[IMaterialVisualPropertiesData Members](#)
[IModelDocExtension::ViewDisplayRealView Property](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

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Scale2 Property (IMaterialVisualPropertiesData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMaterialVisualPropertiesData Interface](#) : Scale2 Property (IMaterialVisualPropertiesData)

Gets or sets the scale factor for the standard texture.

.NET Syntax

Visual Basic (Declaration)

```
Property Scale2 As System.Double
```

Visual Basic (Usage)



```
Dim instance As IMaterialVisualPropertiesData
Dim value As System.Double

instance.Scale2 = value

value = instance.Scale2
```

C#

```
System.double Scale2 {get; set;}
```

C++/CLI

```
property System.double Scale2 {
    System.double get();
    void set ( &    System.double value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Value by which to scale the texture

Visual Basic for Applications (VBA) Syntax

See [MaterialVisualPropertiesData::Scale2](#).

Example

See [IMaterialVisualPropertiesData](#) examples.

Remarks

This property only applies to SOLIDWORKS standard textures.

See Also

[IMaterialVisualPropertiesData Interface](#)
[IMaterialVisualPropertiesData Members](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

IMathPoint (Type)

SOLIDWORKS API Help

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IMathPoint Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : IMathPoint Interface

Provides a simplified interface for manipulating math-point objects' data and ways to create other math objects.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

Visual Basic Syntax

Visual Basic (Declaration)

```
Public Interface IMathPoint
```

Visual Basic (Usage)

```
Dim instance As IMathPoint
```

C#

```
public interface IMathPoint
```

C++/CLI

```
public interface class IMathPoint
```

Visual Basic for Applications (VBA) Syntax

See [MathPoint](#).

Example

[Get Points of Repeating Elements in Table-driven Pattern \(C#\)](#)
[Get Points of Repeating Elements in Table-driven Pattern \(VB.NET\)](#)
[Get Points of Repeating Elements in Table-driven Pattern \(VBA\)](#)

Accessors

[ICamera::GetPosition](#)

[ICamera::GetPositionEntity](#)

[ICamera::GetTargetPointEntity](#)

[ICamera::TargetPointPosition](#)

[IDimension::IGetReferencePoints](#)

[IDragArrowManipulator::Origin](#)

[IFace2::GetProjectedPointOn](#)

[ILibraryFormToolFeatureData::PickPoint](#)

[IMathPoint::AddVector](#) and [IMathPoint::!AddVector](#)

[IMathPoint::MultiplyTransform](#) and [IMathPoint::!MultiplyTransform](#)

[IMathPoint::Scale](#) and [IMathPoint::!Scale](#)

[IMathPoint::SubtractVector](#) and [IMathPoint::!SubtractVector](#)

[IMathUtility::CreatePoint](#) and [IMathUtility::!CreatePoint](#)

[IMathVector::ConvertToPoint](#) and [IMathVector::!ConvertToPoint](#)

[IModelDocExtension::GetVisibleBox](#)

[INote::GetBendLinesValues](#)

[INote::GetTextPoint2](#) and [INote::!GetTextPoint2](#)

[IPlaneManipulator::Origin](#)

- [IRefPlane::BoundingBox](#)
- [IRefPlane::CornerPoints](#)
- [IRefPlane::IGetBoundingBox](#)
- [IRefPlane::IGetCornerPoints](#)
- [IRefPoint::GetRefPoint](#)
- [ISketchBlockDefinition::InsertionPoint](#)
- [ISketchBlockInstance::InsertionPosition](#)
- [ISpring::SectionProfileCenter](#)
- [ISurface::GetProjectedPointOn](#)
- [ISwScene::GetFloorNormal](#)
- [ITriadManipulator::Origin](#)
- [ITriadManipulator::PreviousDragPoint](#)

 Access Diagram

[MathPoint](#)

 See Also

- [IMathPoint Members](#)
- [SolidWorks.Interop.sldworks Namespace](#)
- [IMathTransform Interface](#)
- [IMathUtility Interface](#)
- [IMathVector Interface](#)

IMathPoint Interface Members

[See Also](#) [Properties](#) [Methods](#)


SolidWorks.Interop.sldworks Namespace : IMathPoint Interface

The following tables list the members exposed by [IMathPoint](#).

 Public Properties

Name	Description
 ArrayData	Gets or sets the array of x,y,z coordinates that describe this math point.
 ArrayData	Gets or sets the array of x,y,z coordinates that describe this math point.

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 Public Methods

Name	Description
 AddVector	Translates a math point by a math vector to create a new math point.
 ConvertToVector	Converts a math point to a math vector by using the three coordinates of the math point for the components of the math vector.
 IAddVector	Translates a math point by a math vector to create a new math point.
 IConvertToVector	Converts a math point to a math vector by using the three coordinates of the math point for the components of the math vector.
 IMultiplyTransform	Multiplies a math point with a math transform; the point is rotated, scaled, and then translated.
 IScale	Scales a math point's magnitude.
 ISubtract	Gets a math vector that describes the difference between the math point magnitude from the calling math point.
 ISubtractVector	Gets a math point that describes the difference between a math vector's magnitude from the calling math point
 MultiplyTransform	Multiplies a math point with a math transform; the point is rotated, scaled, and then translated.
 Scale	Scales a math point's magnitude.
 Subtract	Gets a math vector that describes the difference between the math point magnitude from the calling math point.
 SubtractVector	Gets a math point that describes the difference between a math vector's magnitude from the calling math point

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See Also

[IMathPoint Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IMathTransform Interface](#)
[IMathUtility Interface](#)
[IMathVector Interface](#)

SOLIDWORKS API Help

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IMathPoint Interface Methods

[See Also](#)

[SolidWorks.Interop.sldworks Namespace](#) : IMathPoint Interface

For a list of all members of this type, see [IMathPoint members](#).

Public Methods

Name	Description
 AddVector	Translates a math point by a math vector to create a new math point.
 ConvertToVector	Converts a math point to a math vector by using the three coordinates of the math point for the components of the math vector.
 IAddVector	Translates a math point by a math vector to create a new math point.
 IConvertToVector	Converts a math point to a math vector by using the three coordinates of the math point for the components of the math vector.
 IMultiplyTransform	Multiplies a math point with a math transform; the point is rotated, scaled, and then translated.
 IScale	Scales a math point's magnitude.
 ISubtract	Gets a math vector that describes the difference between the math point magnitude from the calling math point.
 ISubtractVector	Gets a math point that describes the difference between a math vector's magnitude from the calling math point
 MultiplyTransform	Multiplies a math point with a math transform; the point is rotated, scaled, and then translated.
 Scale	Scales a math point's magnitude.
 Subtract	Gets a math vector that describes the difference between the math point magnitude from the calling math point.
 SubtractVector	Gets a math point that describes the difference between a math vector's magnitude from the calling math point

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See Also

[IMathPoint Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IMathTransform Interface](#)
[IMathUtility Interface](#)
[IMathVector Interface](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

IMathPoint Interface Properties

[See Also](#)

[SolidWorks.Interop.sldworks Namespace](#) : IMathPoint Interface

For a list of all members of this type, see [IMathPoint members](#).

Public Properties

Name	Description
 ArrayData	Gets or sets the array of x,y,z coordinates that describe this math point.
 IArrayData	Gets or sets the array of x,y,z coordinates that describe this math point.

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See Also

[IMathPoint Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IMathTransform Interface](#)
[IMathUtility Interface](#)
[IMathVector Interface](#)

AddVector Method (IMathPoint)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMathPoint Interface](#) : AddVector Method (IMathPoint)

Translates a math point by a math vector to create a new math point.

.NET Syntax

Visual Basic (Declaration)

```
Function AddVector( _  
    ByVal VectorObjIn As System.Object _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IMathPoint  
Dim VectorObjIn As System.Object  
Dim value As System.Object
```

```
value = instance.AddVector(VectorObjIn)
```

C#

```
System.object AddVector(  
    System.object VectorObjIn  
)
```

C++/CLI

```
System.Object^ AddVector(  
& System.Object^ VectorObjIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

VectorObjIn
[Math vector](#) by which to translate this math point

Return Value

Newly created translated [math point](#) or NULL if the operation fails

Visual Basic for Applications (VBA) Syntax

See [MathPoint::AddVector](#).

Example

[Locate Apex of Conical Face \(VBA\)](#)

See Also

[IMathPoint Interface](#)
[IMathPoint Members](#)
[IMathPoint::IAddVector Method](#)
[IMathPoint::SubtractVector Method](#)
[IMathPoint::ISubtractVector Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

ArrayData Property (IMathPoint)

[See Also](#) [Example](#)



Gets or sets the array of x,y,z coordinates that describe this math point.

.NET Syntax

Visual Basic (Declaration)

```
Property ArrayData As System.Object
```

Visual Basic (Usage)



```
Dim instance As IMathPoint
Dim value As System.Object

instance.ArrayData = value

value = instance.ArrayData
```

C#

```
System.object ArrayData {get; set;}
```

C++/CLI

```
property System.Object^ ArrayData {
    System.Object^ get();
    void set ( & System.Object^ value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Array of 3 doubles representing the x,y,z coordinates of the math point

Visual Basic for Applications (VBA) Syntax

See [MathPoint.ArrayData](#).

Example

[Dimension Edge in Drawing \(VBA\)](#)
[Get Names of Sketch Segments \(VBA\)](#)
[Get Reference Point Data \(VBA\)](#)
[Insert Reference Points \(VBA\)](#)
[Transform Sketch to Model \(VBA\)](#)
[Get Points of Repeating Elements in Table-driven Pattern \(C#\)](#)
[Get Points of Repeating Elements in Table-driven Pattern \(VB.NET\)](#)
[Get Points of Repeating Elements in Table-driven Pattern \(VBA\)](#)
[Get Coordinates of the Plane's Bounding Box \(C#\)](#)
[Get Coordinates of the Plane's Bounding Box \(VB.NET\)](#)
[Get Coordinates of the Plane's Bounding Box \(VBA\)](#)

See Also

[IMathPoint Interface](#)
[IMathPoint Members](#)
[IMathPoint.ArrayData Property](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

ConvertToVector Method (IMathPoint)

[See Also](#)



Converts a math point to a math vector by using the three coordinates of the math point for the components of the math vector.

.NET Syntax

Visual Basic (Declaration)

```
Function ConvertToVector() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IMathPoint  
Dim value As System.Object  
  
value = instance.ConvertToVector()
```

C#

```
System.object ConvertToVector()
```

C++/CLI

```
System.Object^ ConvertToVector();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Newly created [math vector](#) or NULL if the operation fails

Visual Basic for Applications (VBA) Syntax

See [MathPoint::ConvertToVector](#).

See Also

[IMathPoint Interface](#)
[IMathPoint Members](#)
[IMathPoint::IConvertToVector Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

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IAddVector Method (IMathPoint)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMathPoint Interface](#) : IAddVector Method (IMathPoint)

Translates a math point by a math vector to create a new math point.

.NET Syntax

Visual Basic (Declaration)

```
Function IAddVector( _  
    ByVal VectorObjIn As MathVector _  
) As MathPoint
```

Visual Basic (Usage)



```
Dim instance As IMathPoint  
Dim VectorObjIn As MathVector  
Dim value As MathPoint  
  
value = instance.IAddVector(VectorObjIn)
```

C#

```
MathPoint IAddVector(  
    MathVector VectorObjIn  
)
```

C++/CLI


```
MathPoint^ IAddVector(  
& MathVector^ VectorObjIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

VectorObjIn
[Math vector](#) by which to translate this math point

Return Value

Newly created translated [math point](#) or NULL if the operation fails

Visual Basic for Applications (VBA) Syntax

See [MathPoint::IAddVector](#).

See Also

[IMathPoint Interface](#)
[IMathPoint Members](#)
[IMathPoint::AddVector Method](#)
[IMathPoint::SubtractVector Method](#)
[IMathPoint::ISubtractVector Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

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IArrayData Property (IMathPoint)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMathPoint Interface](#) : IArrayData Property (IMathPoint)

Gets or sets the array of x,y,z coordinates that describe this math point.

.NET Syntax

Visual Basic (Declaration)

```
Property IArrayData As System.Double
```

Visual Basic (Usage)



```
Dim instance As IMathPoint  
Dim value As System.Double  
  
instance.IArrayData = value  
  
value = instance.IArrayData
```

C#

```
System.double IArrayData {get; set;}
```

C++/CLI

```
property System.double IArrayData {  
    System.double get();  
    void set ( & System.double value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

- in-process, unmanaged C++: Pointer to an array of 3 doubles representing the x,y,z coordinates of the math point
- VBA, VB.NET, C#, and C++/CLI: Not supported

See Also

[IMathPoint Interface](#)
[IMathPoint Members](#)
[IMathPoint::ArrayData Property](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

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IConvertToVector Method (IMathPoint)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMathPoint Interface](#) : IConvertToVector Method (IMathPoint)

Converts a math point to a math vector by using the three coordinates of the math point for the components of the math vector.

.NET Syntax

Visual Basic (Declaration)

```
Function IConvertToVector() As MathVector
```

Visual Basic (Usage)



```
Dim instance As IMathPoint  
Dim value As MathVector
```

```
value = instance.IConvertToVector()
```

C#

```
MathVector IConvertToVector()
```

C++/CLI

```
MathVector^ IConvertToVector();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

Newly created [math vector](#) or NULL if the operation fails

Visual Basic for Applications (VBA) Syntax

See [MathPoint::IConvertToVector](#).

See Also

[IMathPoint Interface](#)
[IMathPoint Members](#)
[IMathPoint::ConvertToVector Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IMultiplyTransform Method (IMathPoint)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMathPoint Interface](#) : IMultiplyTransform Method (IMathPoint)

Multiplies a math point with a math transform; the point is rotated, scaled, and then translated.

.NET Syntax

Visual Basic (Declaration)

```
Function IMultiplyTransform( _  
    ByVal TransformObjIn As MathTransform _  
) As MathPoint
```

Visual Basic (Usage)



```
Dim instance As IMathPoint  
Dim TransformObjIn As MathTransform  
Dim value As MathPoint  
  
value = instance.IMultiplyTransform(TransformObjIn)
```

C#

```
MathPoint IMultiplyTransform(  
    MathTransform TransformObjIn  
)
```

C++/CLI

```
MathPoint^ IMultiplyTransform(  
& MathTransform^ TransformObjIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TransformObjIn
[Math transform](#) by which to multiply this math point

Return Value

Newly created translated [math point](#) or null if the operation fails

Visual Basic for Applications (VBA) Syntax

See [MathPoint::IMultiplyTransform](#).

See Also

[IMathPoint Interface](#)
[IMathPoint Members](#)
[IMathPoint::MultiplyTransform Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

IScale Method (IMathPoint)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMathPoint Interface](#) : IScale Method (IMathPoint)

Scales a math point's magnitude.

.NET Syntax

Visual Basic (Declaration)

```
Function IScale( _  
    ByVal ValueIn As System.Double _  
) As MathPoint
```

Visual Basic (Usage)



```
Dim instance As IMathPoint  
Dim ValueIn As System.Double
```

```
Dim value As MathPoint

value = instance.IScale(ValueIn)
```

C#

```
MathPoint IScale(
    System.double ValueIn
)
```

C++/CLI

```
MathPoint^ IScale(
    & System.double ValueIn
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ValueIn
Scale by which to multiply the magnitude of the math point

Return Value

Newly created scaled [math point](#) or NULL if the operation fails

Visual Basic for Applications (VBA) Syntax

See [MathPoint::IScale](#).

See Also

[IMathPoint Interface](#)
[IMathPoint Members](#)
[IMathPoint::Scale Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

ISubtract Method (IMathPoint)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMathPoint Interface](#) : ISubtract Method (IMathPoint)

Gets a math vector that describes the difference between the math point magnitude from the calling math point.

.NET Syntax

Visual Basic (Declaration)

```
Function ISubtract( _
    ByVal PointObjIn As MathPoint _
) As MathVector
```

Visual Basic (Usage)



```
Dim instance As IMathPoint
Dim PointObjIn As MathPoint
Dim value As MathVector

value = instance.ISubtract(PointObjIn)
```

C#

```
MathVector ISubtract(
    MathPoint PointObjIn
)
```

C++/CLI

```
MathVector^ ISubtract(  
& MathPoint^ PointObjIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PointObjIn
[Math point](#) by which to subtract this math point

Return Value

Newly created [math vector](#) or NULL if the operation fails

Visual Basic for Applications (VBA) Syntax

See [MathPoint::ISubtract](#).

See Also

[IMathPoint Interface](#)
[IMathPoint Members](#)
[IMathPoint::Subtract Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

ISubtractVector Method (IMathPoint)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMathPoint Interface](#) : ISubtractVector Method (IMathPoint)

Gets a math point that describes the difference between a math vector's magnitude from the calling math point

.NET Syntax

Visual Basic (Declaration)

```
Function ISubtractVector( _  
    ByVal VectorObjIn As MathVector _  
) As MathPoint
```

Visual Basic (Usage)



```
Dim instance As IMathPoint  
Dim VectorObjIn As MathVector  
Dim value As MathPoint  
  
value = instance.ISubtractVector(VectorObjIn)
```

C#

```
MathPoint ISubtractVector(  
    MathVector VectorObjIn  
)
```

C++/CLI

```
MathPoint^ ISubtractVector(  
& MathVector^ VectorObjIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

VectorObjIn
[Math vector](#) by which to subtract this math point

Return Value

Newly created [math point](#) or NULL if the operation fails

Visual Basic for Applications (VBA) Syntax

See [MathPoint::ISubtractVector](#).

See Also

[IMathPoint Interface](#)
[IMathPoint Members](#)
[IMathPoint::SubtractVector Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

MultiplyTransform Method (IMathPoint)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMathPoint Interface](#) : MultiplyTransform Method (IMathPoint)

Multiplies a math point with a math transform; the point is rotated, scaled, and then translated.

.NET Syntax

Visual Basic (Declaration)

```
Function MultiplyTransform( _  
    ByVal TransformObjIn As System.Object _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IMathPoint  
Dim TransformObjIn As System.Object  
Dim value As System.Object  
  
value = instance.MultiplyTransform(TransformObjIn)
```

C#

```
System.object MultiplyTransform(  
    System.object TransformObjIn  
)
```

C++/CLI

```
System.Object^ MultiplyTransform(  
& System.Object^ TransformObjIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TransformObjIn

Math transform by which to multiply this math point

Return Value

Newly created translated [math point](#) or null if the operation fails

Visual Basic for Applications (VBA) Syntax

See [MathPoint::MultiplyTransform](#).

Example

[Dimension Edge in Drawing \(VBA\)](#)
[Evaluate Curves Defined in Sketch Space \(VBA\)](#)
[Get Names of Sketch Segments \(VBA\)](#)
[Transform Coordinates from Sketch to Model Space \(VBA\)](#)
[Transform Sketch to Model \(VBA\)](#)
[Zoom to Region \(VBA\)](#)
[Get Points of Repeating Elements in Table-driven Pattern \(C#\)](#)
[Get Points of Repeating Elements in Table-driven Pattern \(VB.NET\)](#)
[Get Points of Repeating Elements in Table-driven Pattern \(VBA\)](#)

See Also

[IMathPoint Interface](#)
[IMathPoint Members](#)
[IMathPoint::IMultiplyTransform Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

Scale Method (IMathPoint)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMathPoint Interface](#) : Scale Method (IMathPoint)

Scales a math point's magnitude.

.NET Syntax

Visual Basic (Declaration)

```
Function Scale( _  
    ByVal ValueIn As System.Double _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IMathPoint  
Dim ValueIn As System.Double  
Dim value As System.Object  
  
value = instance.Scale(ValueIn)
```

C#

```
System.object Scale(  
    System.double ValueIn  
)
```

C++/CLI

```
System.Object^ Scale(  
&    System.double ValueIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ValueIn

Scale by which to multiply the magnitude of the math point

Return Value

Newly created scaled [math point](#) or NULL if the operation fails

Visual Basic for Applications (VBA) Syntax

See [MathPoint::Scale](#).

See Also

[IMathPoint Interface](#)
[IMathPoint Members](#)
[IMathPoint::IScale Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

Subtract Method (IMathPoint)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMathPoint Interface](#) : Subtract Method (IMathPoint)

Gets a math vector that describes the difference between the math point magnitude from the calling math point.

.NET Syntax

Visual Basic (Declaration)

```
Function Subtract( _  
    ByVal PointObjIn As System.Object _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IMathPoint  
Dim PointObjIn As System.Object  
Dim value As System.Object  
  
value = instance.Subtract(PointObjIn)
```

C#

```
System.object Subtract(  
    System.object PointObjIn  
)
```

C++/CLI

```
System.Object^ Subtract(  
& System.Object^ PointObjIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PointObjIn
[Math point](#) by which to subtract this math point

Return Value

Newly created [math vector](#) or NULL if the operation fails

Visual Basic for Applications (VBA) Syntax

See [MathPoint::Subtract](#).

See Also

[IMathPoint Interface](#)
[IMathPoint Members](#)
[IMathPoint::ISubtract Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SubtractVector Method (IMathPoint)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMathPoint Interface](#) : SubtractVector Method (IMathPoint)

Gets a math point that describes the difference between a math vector's magnitude from the calling math point

.NET Syntax

Visual Basic (Declaration)

```
Function SubtractVector( _  
    ByVal VectorObjIn As System.Object _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IMathPoint  
Dim VectorObjIn As System.Object  
Dim value As System.Object  
  
value = instance.SubtractVector(VectorObjIn)
```

C#

```
System.object SubtractVector(  
    System.object VectorObjIn  
)
```

C++/CLI

```
System.Object^ SubtractVector(  
& System.Object^ VectorObjIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

VectorObjIn

[Math vector](#) by which to subtract this math point

Return Value

Newly created [math point](#) or NULL if the operation fails

Visual Basic for Applications (VBA) Syntax

See [MathPoint::SubtractVector](#).

See Also

[IMathPoint Interface](#)
[IMathPoint Members](#)
[IMathPoint::ISubtractVector Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

IMathTransform (Type)

IMathTransform Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : IMathTransform Interface

Provides a simplified interface for manipulating transformation matrix data.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface IMathTransform
```

Visual Basic (Usage)



```
Dim instance As IMathTransform
```

C#

```
public interface IMathTransform
```

C++/CLI

```
public interface class IMathTransform
```

Visual Basic for Applications (VBA) Syntax

See [MathTransform](#).

Example

[Get Coordinate System Transform \(VBA\)](#)
[Calculate Transformations in Part \(C#\)](#)
[Calculate Transformations in Part \(VB.NET\)](#)
[Calculate Transformations in Part \(VBA\)](#)
[Get Patterned Cosmetic Thread Annotations Data \(C#\)](#)
[Get Patterned Cosmetic Thread Annotations Data \(VB.NET\)](#)
[Get Patterned Cosmetic Thread Annotations Data \(VBA\)](#)

Remarks

Transformation matrix data:

```
| a b c . n |  
| d e f . o |  
| g h i . p |  
| j k l . m |
```

The SOLIDWORKS transformation matrix is stored as a homogeneous matrix of 16 elements, ordered as shown. The first 9 elements (a to i) are elements of a 3x3 rotational sub-matrix, the next 3 elements (j, k, l) define a translation vector, and the next 1 element (m) is a scaling factor. The last 3 elements (n, o, p) are unused in this context.

The 3x3 rotational sub-matrix represents 3 axis sets:

- row 1 for x-axis components of rotation
- row 2 for y-axis components of rotation
- row 3 for z-axis components of rotation

The 3 axes are constrained to be orthogonal and unified so that they produce a pure rotational transformation. Reflections can also be added to these axes by setting the components to negative. The rotation sub-matrix coupled with the lower-left translation vector and the lower-right corner scaling factor creates an affine transformation, which is a transformation that preserves lines and parallelism; i.e., maps parallel lines to parallel lines.

If the 3 axis sets of the 3x3 rotational sub-matrix are not orthogonal or unified, then they are automatically corrected according to the following rules:

- If any axis is 0, or any two axes are parallel, or all axes are coplanar, then an identity matrix replaces the rotational sub-matrix.
- All axes are corrected to be of unit length.

- The axes are built to be orthogonal to each other in the prioritized order of Z, X, Y (X is orthogonal to Z, Y is orthogonal to Z and X).

Accessors

[IAnnotation::GetFipPlaneTransform](#)
[IBody2::GetCoincidenceTransform](#)
[IBody2::GetOriginalPatternedBody](#)
[ICircularPatternFeatureData::GetTransform](#)
[ICollision::GetTransforms](#)
[ICollisionDetectionGroup::GetTransforms](#)
[IComponent2::GetSpecificTransform](#)
[IComponent2::GetTotalTransform](#)
[IComponent2::PresentationTransform](#)
[IComponent2::Transform2](#)
[ICosmeticThreadFeatureData::IGetPatternedTransforms](#)
[ICosmeticThreadFeatureData::PatternedTransforms](#)
[ICThread::IGetPatternedTransforms](#)
[ICThread::PatternedTransforms](#)
[ICurveDrivenPatternFeatureData::GetTransform](#)
[IDerivedPatternFeatureData::GetTransform](#)
[IDisplayDimension::GetDefinitionTransform](#)
[IInterferenceDetectionMgr::GetComponentsAndTransforms](#)
[ILinearPatternFeatureData::GetTransform](#)
[ILocalCircularPatternFeatureData::GetTransform](#)
[ILocalCurvePatternFeatureData::GetTransform](#)
[ILocalLinearPatternFeatureData::GetTransform](#)
[ILocalSketchPatternFeatureData::GetTransform](#)
[IMacroFeatureData::FeatureTransform](#)
[IMacroFeatureData::GetEditTargetTransform](#)
[IMacroFeatureData::GetSelections3](#) and [IMacroFeatureData::IGetSelections3](#)
[IMacroFeatureData::PatternTransform](#)
[IMathTransform::Inverse](#) and [IMathTransform::Inverse](#)
[IMathTransform::Multiply](#) and [IMathTransform::IMultiply](#)
[IMathUtility::ComposeTransform](#)
[IMathUtility::CreateTransform](#) and [IMathUtility::ICreateTransform](#)
[IMathUtility::CreateTransformRotateAxis](#) and [IMathUtility::ICreateTransformRotateAxis](#)
[IMirrorPartFeatureData::GetTransform](#)
[IMirrorSolidFeatureData::GetTransform](#)
[IModelDocExtension::GetCoordinateSystemTransformByName](#)
[IModelView::Orientation3](#)
[IModelView::Transform](#)
[IRefPlane::Transform](#)
[ISectionViewData::GetDynamicCenterTransform](#)
[ISketch::ModelToSketchTransform](#)
[ISketchBlockInstance::BlockToSketchTransform](#)
[ISketchPatternFeatureData::GetTransform](#)
[ITablePatternFeatureData::GetTransform](#)
[IView::ModelToViewTransform](#)
[IView3D::Rotation](#)

Access Diagram

[MathTransform](#)

See Also

[IMathTransform Members](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IMathPoint Interface](#)
[IMathUtility Interface](#)
[IMathVector Interface](#)
[IUserUnit Interface](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IMathTransform Interface Methods

[See Also](#)

[SolidWorks.Interop.sldworks Namespace](#) : IMathTransform Interface

For a list of all members of this type, see [IMathTransform members](#).

Public Methods

Name	Description
 GetData	Obsolete. See IMathTransform.GetData2 .
 GetData2	Gets the math vectors and data that describe the transformation matrix.
 IGetData	Obsolete. See IMathTransform.IGetData2 .
 IGetData2	Gets the math vectors and data that describe the transformation matrix.
 Inverse	Creates a new math transform by inverting the values in an already existing math transform.
 IMultiply	Multiplies two matrices together.
 Inverse	Creates a new math transform by inverting the values in an already existing math transform.
 ISetData	Sets the math vectors and data that describe the transformation matrix.
 Multiply	Multiplies two matrices together.
 SetData	Sets the math vectors and data that describe the transformation matrix.

[Top](#)

See Also

[IMathTransform Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IMathPoint Interface](#)
[IMathUtility Interface](#)
[IMathVector Interface](#)
[IUserUnit Interface](#)

SOLIDWORKS API Help

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IMathTransform Interface Members

[See Also](#) [Properties](#) [Methods](#)

[SolidWorks.Interop.sldworks Namespace](#) : IMathTransform Interface

The following tables list the members exposed by [IMathTransform](#).

Public Properties

Name	Description
 ArrayData	Gets and sets the array of 16 doubles for this transform.
 IArrayData	Gets and sets the array of 16 doubles for this transform.

[Top](#)

Public Methods

Name	Description
GetData	Obsolete. See IMathTransform.GetData2 .

		
	GetData2	Gets the math vectors and data that describe the transformation matrix.
	IGetData	Obsolete. See IMathTransform.IGetData2 .
	IGetData2	Gets the math vectors and data that describe the transformation matrix.
	Inverse	Creates a new math transform by inverting the values in an already existing math transform.
	IMultiply	Multiplies two matrices together.
	Inverse	Creates a new math transform by inverting the values in an already existing math transform.
	ISetData	Sets the math vectors and data that describe the transformation matrix.
	Multiply	Multiplies two matrices together.
	SetData	Sets the math vectors and data that describe the transformation matrix.

[Top](#)

 See Also

[IMathTransform Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IMathPoint Interface](#)
[IMathUtility Interface](#)
[IMathVector Interface](#)
[IUserUnit Interface](#)

SOLIDWORKS API Help

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IMathTransform Interface Properties

[See Also](#)

	
SolidWorks.Interop.sldworks Namespace	: IMathTransform Interface

For a list of all members of this type, see [IMathTransform members](#).

 Public Properties

Name	Description
 ArrayData	Gets and sets the array of 16 doubles for this transform.
 IArrayData	Gets and sets the array of 16 doubles for this transform.

[Top](#)

 See Also

[IMathTransform Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)
[IMathPoint Interface](#)
[IMathUtility Interface](#)
[IMathVector Interface](#)
[IUserUnit Interface](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

ArrayData Property (IMathTransform)

[See Also](#) [Example](#)

		
SolidWorks.Interop.sldworks Namespace	> IMathTransform Interface	: ArrayData Property (IMathTransform)
Gets and sets the array of 16 doubles for this transform.		

 .NET Syntax

Visual Basic (Declaration)

Property ArrayData As System.Object

Visual Basic (Usage)



```
Dim instance As IMathTransform
Dim value As System.Object

instance.ArrayData = value
```

```
value = instance.ArrayData
```

C#

```
System.Object ArrayData {get; set;}
```

C++/CLI

```
property System.Object^ ArrayData {  
    System.Object^ get();  
    void set ( & System.Object^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Array of 16 doubles (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [MathTransform::ArrayData](#).

Example

[Get Angle Between Plane and Line \(VBA\)](#)
[Get Coordinate System Transform \(VBA\)](#)
[Get Transform of Plane \(VBA\)](#)
[Get Transforms of Assembly Components \(VBA\)](#)
[Get Patterned Cosmetic Thread Annotations Data \(C#\)](#)
[Get Patterned Cosmetic Thread Annotations Data \(VB.NET\)](#)
[Get Patterned Cosmetic Thread Annotations Data \(VBA\)](#)

Remarks

The first 9 elements define the 3x3 rotation matrix. The next 3 elements define the translation component. The next element defines the scaling component. The last 3 elements are unused.

The array data argument should be in a form that allows the direct calling of methods such as [IComponent2::Transform2](#).

See Also

[IMathTransform Interface](#)
[IMathTransform Members](#)
[IMathTransform::IArrayData Property](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetData Method (IMathTransform)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMathTransform Interface](#) : GetData Method (IMathTransform)

Obsolete. See [IMathTransform::GetData2](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub GetData(  
    ByRef XAxisObjOut As System.Object, _  
    ByRef YAxisObjOut As System.Object, _  
    ByRef ZAxisObjOut As System.Object, _  
    ByRef TransformObjOut As System.Object, _  
    ByRef ScaleOut As System.Double _  
)
```

Visual Basic (Usage)



```

Dim instance As IMathTransform
Dim XAxisObjOut As System.Object
Dim YAxisObjOut As System.Object
Dim ZAxisObjOut As System.Object
Dim TransformObjOut As System.Object
Dim ScaleOut As System.Double

instance.GetData(XAxisObjOut, YAxisObjOut, ZAxisObjOut, TransformObjOut, ScaleOut)

```

C#

```

void GetData(
    out System.object XAxisObjOut,
    out System.object YAxisObjOut,
    out System.object ZAxisObjOut,
    out System.object TransformObjOut,
    out System.double ScaleOut
)

```

C++/CLI

```

void GetData(
    & [Out] System.Object^ XAxisObjOut,
    & [Out] System.Object^ YAxisObjOut,
    & [Out] System.Object^ ZAxisObjOut,
    & [Out] System.Object^ TransformObjOut,
    & [Out] System.double ScaleOut
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

XAxisObjOut

YAxisObjOut

ZAxisObjOut

TransformObjOut

ScaleOut

Visual Basic for Applications (VBA) Syntax

See [MathTransform::GetData](#).

See Also

[IMathTransform Interface](#)
[IMathTransform Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetData2 Method (IMathTransform)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMathTransform Interface](#) : GetData2 Method (IMathTransform)

Gets the math vectors and data that describe the transformation matrix.

.NET Syntax

Visual Basic (Declaration)

```

Sub GetData2( _
    ByRef XAxisObjOut As System.Object, _
    ByRef YAxisObjOut As System.Object, _
    ByRef ZAxisObjOut As System.Object, _
    ByRef TransformObjOut As System.Object, _
    ByRef ScaleOut As System.Double _
)

```

```

Dim instance As IMathTransform
Dim XAxisObjOut As System.Object
Dim YAxisObjOut As System.Object
Dim ZAxisObjOut As System.Object
Dim TransformObjOut As System.Object
Dim ScaleOut As System.Double

instance.GetData2(XAxisObjOut, YAxisObjOut, ZAxisObjOut, TransformObjOut, ScaleOut)

```

C#

```

void GetData2(
    out System.object XAxisObjOut,
    out System.object YAxisObjOut,
    out System.object ZAxisObjOut,
    out System.object TransformObjOut,
    out System.double ScaleOut
)

```

C++/CLI

```

void GetData2(
    & [Out] System.Object^ XAxisObjOut,
    & [Out] System.Object^ YAxisObjOut,
    & [Out] System.Object^ ZAxisObjOut,
    & [Out] System.Object^ TransformObjOut,
    & [Out] System.double ScaleOut
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

[XAxisObjOut](#)
[Rotation](#) about the x axis
[YAxisObjOut](#)
[Rotation](#) about the y axis
[ZAxisObjOut](#)
[Rotation](#) about the z axis
[TransformObjOut](#)
[Transformation vector](#)
[ScaleOut](#)
Scale

Return Value

The previous version of this method, [IMathTransform::GetData](#), returned inversed x, y, z axes. This version of this method returns the actual x, y, z axes.

Visual Basic for Applications (VBA) Syntax

See [MathTransform::GetData2](#).

See Also

[IMathTransform Interface](#)
[IMathTransform Members](#)
[IMathTransform::GetData2 Method](#)
[IMathTransform::SetData Method](#)
[IMathTransform::SetData Method](#)

Availability

SOLIDWORKS 2005 SP2, Revision Number 13.2

IArrayData Property (IMathTransform)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMathTransform Interface](#) : IArrayData Property (IMathTransform)

Gets and sets the array of 16 doubles for this transform.

Visual Basic Syntax

Visual Basic (Declaration)

```
Property IArrayData As System.Double
```

Visual Basic (Usage)



```
Dim instance As IMathTransform  
Dim value As System.Double
```

```
instance.IArrayData = value
```

```
value = instance.IArrayData
```

C#

```
System.double IArrayData {get; set;}
```

C++/CLI

```
property System.double IArrayData {  
    System.double get();  
    void set ( & System.double value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

- in-process, unmanaged C++: Pointer to an array of 16 doubles (see **Remarks**)
- VBA, VB.NET, C#, and C++/CLI: Not supported

Remarks

The first 9 elements define the 3x3 rotation matrix. The next 3 elements define the translation component. The next element defines the scaling component. The last 3 elements are unused.

The array data argument should be in a form that allows the direct calling of methods such as [IComponent2::Transform2](#).

See Also

[IMathTransform Interface](#)
[IMathTransform Members](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

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IGetData Method (IMathTransform)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMathTransform Interface](#) : IGetData Method (IMathTransform)

Obsolete. See [IMathTransform::IGetData2](#).

Visual Basic Syntax

Visual Basic (Declaration)

```
Sub IGetData( _  
    ByRef XAxisObjOut As MathVector, _  
    ByRef YAxisObjOut As MathVector, _  
    ByRef ZAxisObjOut As MathVector, _  
    ByRef TransformObjOut As MathVector, _  
    ByRef ScaleOut As System.Double _  
)
```

Visual Basic (Usage)



```

Dim instance As IMathTransform
Dim XAxisObjOut As MathVector
Dim YAxisObjOut As MathVector
Dim ZAxisObjOut As MathVector
Dim TransformObjOut As MathVector
Dim ScaleOut As System.Double

instance.IGetData(XAxisObjOut, YAxisObjOut, ZAxisObjOut, TransformObjOut, ScaleOut)

```

C#

```

void IGetData(
    out MathVector XAxisObjOut,
    out MathVector YAxisObjOut,
    out MathVector ZAxisObjOut,
    out MathVector TransformObjOut,
    out System.double ScaleOut
)

```

C++/CLI

```

void IGetData(
    & [Out] MathVector^ XAxisObjOut,
    & [Out] MathVector^ YAxisObjOut,
    & [Out] MathVector^ ZAxisObjOut,
    & [Out] MathVector^ TransformObjOut,
    & [Out] System.double ScaleOut
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

XAxisObjOut

YAxisObjOut

ZAxisObjOut

TransformObjOut

ScaleOut

Visual Basic for Applications (VBA) Syntax

See [MathTransform.IGetData](#).

See Also

[IMathTransform Interface](#)
[IMathTransform Members](#)

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IGetData2 Method (IMathTransform)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMathTransform Interface](#) : IGetData2 Method (IMathTransform)

Gets the math vectors and data that describe the transformation matrix.

.NET Syntax

Visual Basic (Declaration)

```

Sub IGetData2( _
    ByRef XAxisObjOut As MathVector, _
    ByRef YAxisObjOut As MathVector, _
    ByRef ZAxisObjOut As MathVector, _
    ByRef TransformObjOut As MathVector, _
    ByRef ScaleOut As System.Double _
)

```

```

Dim instance As IMathTransform
Dim XAxisObjOut As MathVector
Dim YAxisObjOut As MathVector
Dim ZAxisObjOut As MathVector
Dim TransformObjOut As MathVector
Dim ScaleOut As System.Double

instance.IGetData2(XAxisObjOut, YAxisObjOut, ZAxisObjOut, TransformObjOut, ScaleOut)

```

C#

```

void IGetData2(
    out MathVector XAxisObjOut,
    out MathVector YAxisObjOut,
    out MathVector ZAxisObjOut,
    out MathVector TransformObjOut,
    out System.double ScaleOut
)

```

C++/CLI

```

void IGetData2(
    & [Out] MathVector^ XAxisObjOut,
    & [Out] MathVector^ YAxisObjOut,
    & [Out] MathVector^ ZAxisObjOut,
    & [Out] MathVector^ TransformObjOut,
    & [Out] System.double ScaleOut
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

[XAxisObjOut](#)
[Rotation](#) about the x axis
[YAxisObjOut](#)
[Rotation](#) about the y axis
[ZAxisObjOut](#)
[Rotation](#) about the z axis
[TransformObjOut](#)
[Transformation vector](#)
[ScaleOut](#)
Scale

Return Value

The previous version of this method, [IMathTransform::IGetData](#), returned inversed x, y, z axes. This version of this method returns the actual x, y, z axes.

Visual Basic for Applications (VBA) Syntax

See [MathTransform::IGetData2](#).

See Also

[IMathTransform Interface](#)
[IMathTransform Members](#)
[IMathTransform::GetData2 Method](#)
[IMathTransform::SetData Method](#)
[IMathTransform::SetData Method](#)

Availability

SOLIDWORKS 2005 SP2, Revision 13.2

Inverse Method (IMathTransform)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMathTransform Interface](#) : Inverse Method (IMathTransform)

Creates a new math transform by inverting the values in an already existing math transform.

.NET Syntax

Visual Basic (Declaration)

```
Function IInverse() As MathTransform
```

Visual Basic (Usage)



```
Dim instance As IMathTransform  
Dim value As MathTransform  
  
value = instance.IInverse()
```

C#

```
MathTransform IInverse()
```

C++/CLI

```
MathTransform^ IInverse();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[Math transform](#) or NULL if the operation fails

Visual Basic for Applications (VBA) Syntax

See [MathTransform::IInverse](#).

See Also

[IMathTransform Interface](#)
[IMathTransform Members](#)
[IMathTransform::Inverse Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

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IMultiply Method (IMathTransform)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMathTransform Interface](#) : IMultiply Method (IMathTransform)

Multiplies two matrices together.

.NET Syntax

Visual Basic (Declaration)

```
Function IMultiply( _  
    ByVal TransformObjIn As MathTransform _  
) As MathTransform
```

Visual Basic (Usage)



```
Dim instance As IMathTransform  
Dim TransformObjIn As MathTransform  
Dim value As MathTransform  
  
value = instance.IMultiply(TransformObjIn)
```

C#

```
MathTransform IMultiply(  
    MathTransform TransformObjIn
```

)

C++/CLI

```
MathTransform^ IMultiply(  
& MathTransform^ TransformObjIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TransformObjIn

[Math transform](#) by which to multiply the calling math transform

Return Value

Newly created [math transform](#) object or null if the operation fails

Visual Basic for Applications (VBA) Syntax

See [MathTransform::IMultiply](#).

Remarks

The resulting transform is the result of transforming math transform with respect to the transformObjIn coordinate frame.

See Also

[IMathTransform Interface](#)
[IMathTransform Members](#)
[Multiply Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

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Inverse Method (IMathTransform)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMathTransform Interface](#) : Inverse Method (IMathTransform)

Creates a new math transform by inverting the values in an already existing math transform.

.NET Syntax

Visual Basic (Declaration)

```
Function Inverse() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IMathTransform  
Dim value As System.Object
```

```
value = instance.Inverse()
```

C#

```
System.object Inverse()
```

C++/CLI

```
System.Object^ Inverse();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[Math transform](#) or NULL if the operation fails

Visual Basic for Applications (VBA) Syntax

See [MathTransform::Inverse](#).

Example

[Create Holes Using Hole Wizard and Sketch Points \(VBA\)](#)
[Evaluate Curves Defined in Sketch Space \(VBA\)](#)
[Get Names of Sketch Segments \(VBA\)](#)
[Get x,y,z Locations of Points in Sketch \(VBA\)](#)
[Transform Coordinates from Sketch to Model Space \(VBA\)](#)
[Transform Sketch to Model \(VBA\)](#)
[Get Points of Repeating Elements in Table-driven Pattern \(C#\)](#)
[Get Points of Repeating Elements in Table-driven Pattern \(VB.NET\)](#)
[Get Points of Repeating Elements in Table-driven Pattern \(VBA\)](#)

See Also

[IMathTransform Interface](#)
[IMathTransform Members](#)
[IMathTransform::Inverse Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

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ISetData Method (IMathTransform)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMathTransform Interface](#) : ISetData Method (IMathTransform)

Sets the math vectors and data that describe the transformation matrix.

.NET Syntax

Visual Basic (Declaration)

```
Sub ISetData( _  
    ByVal XAxisObjIn As MathVector, _  
    ByVal YAxisObjIn As MathVector, _  
    ByVal ZAxisObjIn As MathVector, _  
    ByVal TransformObjIn As MathVector, _  
    ByVal ScaleValIn As System.Double _  
)
```

Visual Basic (Usage)



```
Dim instance As IMathTransform  
Dim XAxisObjIn As MathVector  
Dim YAxisObjIn As MathVector  
Dim ZAxisObjIn As MathVector  
Dim TransformObjIn As MathVector  
Dim ScaleValIn As System.Double  
  
instance.ISetData(XAxisObjIn, YAxisObjIn, ZAxisObjIn, TransformObjIn, ScaleValIn)
```

C#

```
void ISetData(  
    MathVector XAxisObjIn,  
    MathVector YAxisObjIn,  
    MathVector ZAxisObjIn,  
    MathVector TransformObjIn,  
    System.double ScaleValIn  
)
```

C++/CLI

```
void ISetData(  
& MathVector^ XAxisObjIn,
```

```

& MathVector^ YAxisObjIn,
& MathVector^ ZAxisObjIn,
& MathVector^ TransformObjIn,
& System.double ScaleValIn
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

[XAxisObjIn](#)
 X rotation [math vector](#) object of the transform
[YAxisObjIn](#)
 Y rotation [math vector](#) object of the transform
[ZAxisObjIn](#)
 Z rotation [math vector](#) object of the transform
[TransformObjIn](#)
 Translation [math vector](#) object of the transform
[ScaleValIn](#)
 Scale component of the transform

Visual Basic for Applications (VBA) Syntax

See [MathTransform::ISetData](#).

See Also

[IMathTransform Interface](#)
[IMathTransform Members](#)
[IMathTransform::SetData Method](#)
[IMathTransform::GetData2 Method](#)
[IMathTransform::IGetData2 Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

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Multiply Method (IMathTransform)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMathTransform Interface](#) : Multiply Method (IMathTransform)

Multiplies two matrices together.

.NET Syntax

Visual Basic (Declaration)

```

Function Multiply(
    ByVal TransformObjIn As System.Object _
) As System.Object

```

Visual Basic (Usage)



```

Dim instance As IMathTransform
Dim TransformObjIn As System.Object
Dim value As System.Object

value = instance.Multiply(TransformObjIn)

```

C#

```

System.object Multiply(
    System.object TransformObjIn
)

```

C++/CLI

```

System.Object^ Multiply(
    & System.Object^ TransformObjIn
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TransformObjIn

[Math transform](#) by which to multiply the calling math transform

Return Value

Newly created [math transform](#) object or null or Nothing if the operation fails

Visual Basic for Applications (VBA) Syntax

See [MathTransform::Multiply](#).

Example

[Set Components and Transforms for Interference Detection \(C#\)](#)
[Set Components and Transforms for Interference Detection \(VB.NET\)](#)
[Set Components and Transforms for Interference Detection \(VBA\)](#)

Remarks

The resulting transform is the result of transforming the math transform with respect to the transformObjIn coordinate frame.

See Also

[IMathTransform Interface](#)
[IMathTransform Members](#)
[IMathTransform::IMultiply Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

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SetData Method (IMathTransform)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IMathTransform Interface](#) : SetData Method (IMathTransform)

Sets the math vectors and data that describe the transformation matrix.

.NET Syntax

Visual Basic (Declaration)

```
Sub SetData( _  
    ByVal XAxisObjIn As System.Object, _  
    ByVal YAxisObjIn As System.Object, _  
    ByVal ZAxisObjIn As System.Object, _  
    ByVal TransformObjIn As System.Object, _  
    ByVal ScaleValIn As System.Double _  
)
```

Visual Basic (Usage)

```
Dim instance As IMathTransform  
Dim XAxisObjIn As System.Object  
Dim YAxisObjIn As System.Object  
Dim ZAxisObjIn As System.Object  
Dim TransformObjIn As System.Object  
Dim ScaleValIn As System.Double  
  
instance.SetData(XAxisObjIn, YAxisObjIn, ZAxisObjIn, TransformObjIn, ScaleValIn)
```

C#

```
void SetData(  
    System.object XAxisObjIn,  
    System.object YAxisObjIn,
```



Copy Code


```
System.object ZAxisObjIn,  
System.object TransformObjIn,  
System.double ScaleValIn  
)
```

C++/CLI

```
void SetData(  
& System.Object^ XAxisObjIn,  
& System.Object^ YAxisObjIn,  
& System.Object^ ZAxisObjIn,  
& System.Object^ TransformObjIn,  
& System.double ScaleValIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

XAxisObjIn
X rotation [math vector](#) object of the transform
YAxisObjIn
Y rotation [math vector](#) object of the transform
ZAxisObjIn
Z rotation [math vector](#) object of the transform
TransformObjIn
Translation [math vector](#) object of the transform
ScaleValIn
Scale component of the transform

Visual Basic for Applications (VBA) Syntax

See [MathTransform::SetData](#).

See Also

[IMathTransform Interface](#)
[IMathTransform Members](#)
[IMathTransform::SetData Method](#)
[IMathTransform::GetData2 Method](#)
[IMathTransform::GetData2 Method](#)

Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0